

Activity 2: AI-Assisted Pull Request Review

Purpose

Learn how to use AI to analyze Pull Request (PR) changes, identify potential issues, assess code quality, and provide meaningful review feedback before code is approved for merge.

Pre-requisites

- Completion of Activity 1
- Working SmartBank EMI Calculator application
- VS Code with Cline configured

Scenario

A fellow developer has submitted a Pull Request (PR) with enhancements to the EMI Calculator. The proposed changes include improved input validation, UI updates, and code refactoring.

As a reviewer, your responsibility is to use AI to analyze the proposed changes, identify potential risks, verify coding standards, and recommend whether the Pull Request is ready for approval.

Tasks

Task 1: Analyze the Pull Request Changes

Review the modified code and understand what has changed.

Suggested Prompt

Act as a Senior Software Engineer reviewing a Pull Request for the Oracle JET EMI Calculator.

Analyze the proposed code changes.

Identify:

- Features added
- Files modified
- Business logic changes
- UI changes
- Validation changes
- Refactoring performed

Summarize the Pull Request in simple language suitable for a reviewer.

Do not modify the code.

Task 2: Review Code Quality

Evaluate whether the changes follow coding standards and enterprise best practices.

Suggested Prompt

Review the Pull Request for code quality.

Evaluate:

- Readability
- Maintainability
- Naming conventions
- Code duplication
- Modularity
- Error handling
- Oracle JET and KnockoutJS best practices

Identify any issues that should be resolved before merging.

Do not rewrite the code.

Task 3: Identify Risks and Missing Scenarios

Determine whether the proposed changes introduce any risks or require additional validation.

Suggested Prompt

Review the Pull Request from a quality assurance perspective.

Identify:

- Potential bugs
- Missing validation
- Edge cases
- Boundary conditions
- Performance concerns
- Security considerations
- User experience issues

Explain why each issue should be addressed before approving the Pull Request.

Task 4: Generate Pull Request Review Comments

Prepare professional review comments that could be shared with the developer.

Suggested Prompt

Act as a Senior Code Reviewer.

Generate professional Pull Request review comments.

For each issue identified:

- Mention the affected area.
- Describe the concern.
- Explain why it should be improved.
- Suggest a practical recommendation.

Use a constructive and professional review style suitable for an enterprise development team.

Expected Deliverables

By the end of this activity, you should have:

- Pull Request Summary
- Code Quality Review
- Risk Assessment Report
- Professional Pull Request Review Comments